

LEARNING RESOURCE GUIDE

ISC2 CCSP: Certified Cloud Security Professional (Aug 2026) | Professional Participant Workbook

Learning Objectives

By the end of this lecture and its accompanying guide, you will be able to:

- By the end of this lecture you will be able to define cloud shared considerations and explain why they function as a structured evaluation framework rather than a technical checklist.
- By the end of this lecture you will be able to distinguish portability, interoperability, and reversibility as distinct concerns and identify the security risks associated with weaknesses in each.
- By the end of this lecture you will be able to explain the security dimensions of availability, performance, and resiliency as shared considerations in cloud procurement.
- By the end of this lecture you will be able to describe the role of security and privacy as contractual and operational shared considerations, identifying the questions that must be answered before service commencement.
- By the end of this lecture you will be able to evaluate a cloud Service Level Agreement for adequacy, identifying provisions that protect the customer and exclusions that weaken enforceability.
- By the end of this lecture you will be able to apply shared considerations as a structured procurement evaluation process, mapping provider offerings against organisational requirements and identifying gaps.

Introduction

In 2019, a major financial institution completed a multi-year migration of its core transaction processing to a public cloud provider. The migration was technically successful. Three months after go-live, the organisation discovered that a clause in their cloud contract meant the provider could retain copies of processed data for up to eighteen months after termination of the agreement. The legal team had not read that clause carefully. The security team had not flagged it. Nobody had mapped the data flows carefully enough to understand what reversibility meant in practice when that relationship ended. This incident illustrates exactly why cloud shared considerations exist as a formal topic in the CCSP exam outline and as a discipline in professional cloud security practice.

Shared considerations are the set of concerns that every organisation must think through when evaluating, procuring, or operating a cloud service. They sit above the purely technical layer. They are the questions you ask before you sign a contract, the obligations you track while you are operating, and the conditions you verify when you are planning to leave. The first cluster of shared considerations covered in this lecture focuses on how services work together, how you move in and out of them, and what level of service you can rely on: portability, interoperability, reversibility, availability, security, privacy, resiliency, performance, and the governance of these commitments through Service Level Agreements.

The financial institution story is not unusual. Organisations regularly discover post-procurement that their assumptions about cloud relationships differed materially from the contractual reality. The purpose of shared

considerations as a discipline is to prevent that discovery from happening at the worst possible moment. A practitioner who treats shared considerations as a structured pre-procurement exercise, rather than a post-incident retrospective, protects the organisation from the class of risks that the financial institution experienced.

Core Concepts Explained

Portability

Definition: The ability to move data, applications, and workloads from one cloud environment to another without prohibitive cost, technical barrier, or loss of functionality. A cloud service with poor portability traps customers in proprietary formats or technologies that make migration economically or technically impractical.

Business relevance: Portability is a strategic business requirement because it preserves the organisation's negotiating leverage with cloud providers and its ability to respond to changes in provider pricing, service quality, compliance posture, or security posture. A provider that knows you cannot easily leave has less incentive to maintain competitive terms.

Cloud Security Practitioner relevance: A Cloud Security Practitioner must evaluate portability during procurement by assessing whether data can be exported in standard formats, whether application dependencies on proprietary services would need to be re-engineered for migration, and whether the contract specifies portability obligations including export mechanisms and timelines.

Real-world example: A US media company migrated its content delivery pipeline to a proprietary cloud-native streaming service in 2018, taking advantage of superior transcoding performance. By 2022, the provider had increased licensing costs by 40% over three years. When the organisation evaluated migration to an alternative provider, the engineering team estimated that re-engineering the pipeline away from the provider's proprietary transcoding API would require fourteen months of development effort and cost more than three years of the price increase. The organisation had effectively forfeited its negotiating leverage by adopting a service without evaluating its portability implications. The provider's standard contract had contained no portability guarantee, and the organisation had not negotiated one. Portability should have been assessed as a risk factor before the initial adoption decision.

Risks or limitations: Deep dependency on proprietary data formats or services reduces portability to near zero without significant re-engineering. The best practice is to evaluate portability during procurement by testing data export in standard formats and assessing the re-engineering effort required to remove provider-specific dependencies.

Interoperability

Definition: The ability of different cloud services and systems to work together effectively, exchanging data and functions across provider boundaries. A service that only integrates cleanly with other services from the same provider may be deeply interoperable within that ecosystem but unable to connect to anything outside it.

Business relevance: Interoperability directly affects the organisation's ability to connect cloud services to existing on-premises systems, partner systems, and regulatory reporting platforms. Poor interoperability creates integration costs, data silos, and manual bridging processes that introduce both operational inefficiency and security risk through non-standard data handling.

Cloud Security Practitioner relevance: A Cloud Security Practitioner evaluating interoperability must assess whether services use standard APIs and protocols, whether authentication mechanisms such as SAML, OAuth, or OpenID Connect are supported for federated identity, and whether data exchange formats such as JSON, XML, or HL7 for healthcare are compatible with existing systems and regulatory reporting requirements.

Real-world example: A UK National Health Service trust selected a cloud-based clinical documentation system whose API was proprietary and undocumented for external consumption. The trust's existing patient administration system, which managed appointment scheduling and bed management, could not integrate with the new clinical documentation platform because the documentation system's vendor had not published an interoperability interface. Patient data had to be manually transcribed between systems for six months while a bespoke integration was developed at significant cost. The integration itself introduced a security risk because it required storing credentials for both systems in a shared configuration file on an intermediate server, a pattern that was identified as a vulnerability in a subsequent penetration test. Interoperability should have been a hard requirement in the procurement specification.

Risks or limitations: Proprietary APIs and non-standard protocols create integration gaps that are often bridged by insecure workarounds. The best practice is to require published, versioned, standard-protocol APIs as a procurement prerequisite and to test integration against existing systems before contract signature.

Reversibility

Definition: The ability to recover data completely from a cloud service and terminate the relationship cleanly, including the provider ceasing to hold copies of the customer's data after a defined period. Sometimes called exit strategy. Poor reversibility leaves data in the provider's possession beyond the intended relationship.

Business relevance: Reversibility is a risk management requirement because it determines what happens when the organisation needs to exit a cloud relationship. Exits can be planned, such as a strategic migration, or forced, such as a provider compliance failure, security incident, or business failure. In either case, the organisation must be able to recover its data and verify that the provider no longer holds it.

Cloud Security Practitioner relevance: A Cloud Security Practitioner must review exit clauses in cloud contracts before signature, specifically verifying what happens to data after termination, how long the provider retains copies, in what format data is returned, what the timeline for complete data deletion is, and whether the provider will certify deletion. The financial institution in this lecture's opening story discovered these terms too late.

Real-world example: The financial institution described in the lecture opening migrated transaction processing to a public cloud and discovered three months after go-live that the provider's standard contract permitted retention of processed data copies for eighteen months after termination. The legal and security teams had not reviewed the exit clause during procurement. When the organisation's legal team finally read the clause during a routine contract audit triggered by a potential acquisition, they found no mechanism to request early deletion and no certification process for data destruction. Renegotiating the clause after go-live cost three months of legal effort and required the organisation to accept a higher service tier to obtain contract terms that included a data deletion certification process. The corrective practice: exit clause review must be a gate in the procurement process, not a post-go-live activity.

Risks or limitations: Contracts that lack explicit data deletion obligations, timelines, and certification processes leave the organisation's data in the provider's control for indeterminate periods after the relationship ends. This creates privacy, regulatory, and competitive intelligence exposure. The best practice is to negotiate specific deletion timelines, export format obligations, and written deletion certification as contract requirements.

Availability and Service Level Agreements

Definition: Availability describes the committed service uptime expressed as a percentage of time over a measurement period. Service Level Agreements formalise availability targets and other performance

commitments into enforceable contractual terms, specifying measurement methodology, remedy processes, and exclusions.

Business relevance: A published availability figure of 99.9% permits more than eight hours of downtime per year. Whether that is acceptable depends on the business requirements of the workload. The commercial terms of an SLA, including remedies, exclusions, and measurement methodology, determine whether the published figure translates into a real contractual obligation or a marketing aspiration.

Cloud Security Practitioner relevance: A Cloud Security Practitioner reviewing an SLA must assess not only the availability percentage but how availability is measured, what counts as downtime, what exclusions apply such as scheduled maintenance or provider-defined force majeure, what the remedy is when availability falls short, and whether the remedy such as service credits actually compensates the customer for the business impact of the outage.

Real-world example: A logistics company's SLA with its cloud provider specified 99.95% monthly availability. During a major outage affecting an entire AWS region in December 2021, the company experienced fourteen hours of downtime for its order management system during peak season. When the company submitted a compensation claim, the provider applied an exclusion clause covering force majeure events and infrastructure failures beyond a single availability zone, which reduced the applicable downtime to three hours under the contract's measurement methodology. The credit issued was equivalent to three days of service fees, representing a small fraction of the actual business loss from missed deliveries and manual order processing costs. The lesson: negotiate SLA remedies that include business impact caps, not just service credits, and review exclusion clauses specifically for the scenarios most likely to affect your workload.

Risks or limitations: Default SLA terms from cloud providers are written to minimise the provider's financial exposure, not to compensate the customer for actual business impact. The best practice is to negotiate SLA terms, review all exclusion clauses, and ensure that the remedy structure provides meaningful compensation for the availability targets that matter to the business.

Security, Privacy, and Resiliency as Shared Considerations

Definition: Security and privacy are formal shared considerations because they are contractual and operational commitments that must be agreed before service commencement, maintained throughout the relationship, and verified through independent audit. Resiliency is the service's engineering capacity to absorb disruption and continue operating or recover quickly, which enables the availability target to be met under adverse conditions.

Business relevance: Security and privacy shared considerations define the allocation of responsibility between customer and provider. Who is responsible for each control? Who holds what data? In which jurisdictions? What rights does the provider have over customer data if the provider is acquired, compelled by government, or suffers its own breach? These questions must have explicit contractual answers before data is placed with a provider.

Cloud Security Practitioner relevance: A Cloud Security Practitioner evaluating security and privacy as shared considerations must review the provider's data processing agreement, shared responsibility matrix, sub-processor list, breach notification obligations, government data access policies, and audit rights. Resiliency assessment should cover the provider's architecture for multi-data-centre redundancy and automated failover, and whether degraded-mode operation protects data integrity during partial failure.

Real-world example: A UK financial regulator required that regulated firms demonstrate their cloud providers had contractually committed to specific breach notification timelines aligned with UK GDPR requirements, namely notification to the firm within 24 hours of a confirmed breach. Several major cloud providers' standard contracts specified notification within 72 hours, aligned with the EU standard but not the firm-facing

notification obligation the regulated firm held to the regulator. The regulated firm had to negotiate a supplemental data processing addendum with each provider to obtain 24-hour notification obligations. Without this review, the firm would have faced a regulatory enforcement risk in the event of a provider-side breach because it could not have met its own notification deadline if the provider's contractual obligation permitted a delay of up to 72 hours. Security shared considerations, negotiated in advance, prevented a foreseeable compliance failure.

Risks or limitations: Default provider contracts often contain breach notification timelines, data access rights, and sub-processor terms that conflict with the customer's own regulatory obligations. The best practice is to map provider contract terms against every applicable regulatory notification and data protection obligation before service commencement and negotiate supplemental terms where gaps exist.

Practical Applications

The following scenarios show how shared considerations analysis translates into concrete procurement actions and organisational protections across real cloud adoption decisions.

Conducting a Structured Shared Considerations Evaluation During Cloud Procurement

A financial services firm is evaluating three cloud providers for its customer data analytics platform. A Cloud Security Practitioner leads a structured shared considerations evaluation using each consideration as a separate assessment dimension. For portability, the practitioner requests a data export demonstration from each provider in standard formats and documents the re-engineering effort that would be required to move the analytics workload between providers. For interoperability, the practitioner maps the providers' APIs against the firm's existing data warehouse and regulatory reporting system interfaces. For reversibility, the practitioner requests the exit clause language from each provider's standard contract and flags the one that does not specify a data deletion timeline or certification process. For availability, the practitioner reviews SLA terms including exclusion clauses and calculates the effective availability commitment after exclusions are applied. The evaluation produces a gap matrix that maps each provider's offering against the firm's requirements, which becomes the basis for contract negotiation with the preferred provider. Three gaps are identified that require negotiated contract terms before the firm will commit to the service.

Negotiating an SLA That Provides Genuine Protection

A cloud security practitioner is supporting contract negotiation for a healthcare organisation selecting a cloud provider for its patient scheduling system. The provider's standard SLA specifies 99.9% monthly availability with service credit remedies capped at one month's fees. The practitioner calculates that the provider's standard remedy would cover less than two percent of the organisation's estimated business cost from a twelve-hour outage during a peak scheduling period. The practitioner recommends negotiating three specific changes: removal of the scheduled maintenance exclusion from downtime calculation during business hours, replacement of the service credit cap with a remedy structure that scales with actual downtime duration, and addition of an availability guarantee that applies at the application layer rather than the infrastructure layer. The provider accepts two of the three changes. The practitioner documents the remaining gap in the organisation's risk register, noting that the organisation accepts the residual risk and has implemented a compensating control in the form of a manual scheduling backup procedure that can be activated within four hours of a confirmed outage.

Applying Security and Privacy Shared Considerations to a Breach Notification Gap

A cloud security practitioner at a regulated financial firm discovers during a contract review that the firm's primary cloud provider's data processing agreement specifies breach notification to the customer within 72 hours of a confirmed breach. The firm's own obligation under its financial services regulator requires it to notify the regulator within 24 hours of becoming aware of a significant incident. The practitioner identifies a foreseeable scenario in which the provider experiences a breach on a Monday, confirms it on Wednesday, notifies the firm on Friday evening, and the firm receives the notification on Saturday, at which point the 24-hour notification window to the regulator has already passed. The practitioner drafts a supplemental data processing addendum requiring the provider to notify the firm within 12 hours of confirming a breach, allowing the firm time to assess significance and meet its regulatory notification window even if the notification is received outside business hours. The provider accepts the modified timeline after legal review. The practitioner also adds a quarterly review of the provider's sub-processor list to the firm's governance calendar, having identified that the standard contract permitted addition of new sub-processors without prior customer notification.

Best Practices

- **Test data portability before contract signature by requesting a full data export in the formats specified in the contract.** Provider documentation may describe portability capabilities that do not work as described for specific data types or at scale. A portability test using a representative data sample before commitment identifies gaps when there is still time to negotiate or select an alternative provider. Document the test results, including export format, completeness, and time required, as part of the procurement record.
- **Review every SLA exclusion clause for scenarios most likely to affect your specific workload.** Exclusion clauses in cloud SLAs commonly cover scheduled maintenance, force majeure events, customer-caused outages, and infrastructure failures beyond a defined scope. The most consequential exclusions are those that apply to the scenarios most likely to cause downtime for your workload. A workload hosted in a single region is most vulnerable to regional infrastructure failures, which are commonly excluded. Identify the applicable exclusions and negotiate their removal or limitation where the business impact of an uncovered outage would be significant.
- **Negotiate breach notification timelines that align with your most demanding regulatory notification obligation.** Provider standard contracts are written to the minimum legally required notification timeline, which may not align with the customer's own notification obligations to regulators, customers, or partners. Map every applicable notification obligation before reviewing the provider's contract terms and negotiate notification timelines that allow you to meet your most demanding obligation even when notification is received outside business hours.
- **Review and negotiate exit clauses as a condition of initial contract signature, not after go-live.** Exit clause review post-procurement leaves the organisation in a position where renegotiation is expensive and the provider has reduced incentive to offer favourable terms. The exit clause should specify the data export format and timeline, the data deletion timeline and process, a written deletion certification, and the obligations applicable to sub-processors that may hold copies of customer data.
- **Document the shared responsibility matrix for each cloud service and verify it is understood by the teams operating the service.** The shared responsibility model allocates specific security controls between provider and customer, and the allocation differs by service type. A gap in understanding of which party is responsible for a specific control commonly results in that control being implemented by neither party. The shared responsibility matrix should be reviewed with both the security team and the operational team

responsible for each cloud service, and gaps in coverage identified and remediated before the service handles production data.

Common Mistakes and Misconceptions

Misconception: Accepting default cloud contract terms without reviewing exit clauses, assuming the provider relationship will be permanent or that exit can be addressed later.

Correct approach: Exit clause review must be a mandatory gate in the procurement process. The practitioner should verify data deletion timelines, export format obligations, sub-processor deletion requirements, and written certification processes before contract signature.

Misconception: Treating an SLA's published availability percentage as the effective contractual commitment without reviewing exclusion clauses.

Correct approach: The effective availability commitment after applying exclusion clauses may be substantially lower than the headline figure. Every exclusion clause must be reviewed for its applicability to the specific workload, and exclusions covering the most likely downtime scenarios should be negotiated before signature.

Misconception: Assuming that security and privacy shared considerations are addressed by the provider's existing compliance certifications such as ISO 27001 or SOC 2.

Correct approach: Provider certifications confirm that the provider's own controls meet specific standards within a defined scope. They do not address the allocation of responsibility between provider and customer, the customer's specific regulatory obligations, or contractual terms such as breach notification timelines. These must be reviewed and negotiated independently of certification status.

Misconception: Evaluating interoperability only at the point of initial integration, without assessing the impact of provider API versioning and deprecation on future compatibility.

Correct approach: Cloud provider APIs evolve. An integration that works at procurement may break when a provider deprecates an API version. Interoperability evaluation should include review of the provider's API versioning policy, deprecation timelines, and the customer's ability to control which API version their systems use.

table: Cloud Shared Considerations Procurement Evaluation Matrix

Use this matrix to evaluate each shared consideration for a cloud service under procurement. For each consideration, document the provider's offering, the organisational requirement, whether there is a gap, and the required contract negotiation.

Consideration	Provider Offering	Org Requirement	Gap?	Contract Action Required
Portability	Data export in CSV only; no API export	Standard format export (JSON/XML)	Yes	Negotiate export format clause
Interoperability	Proprietary REST API; no SAML federation	SAML 2.0 for federated identity	Yes	Require SAML support or reject service
Reversibility	Data retained 18 months post-termination	Deletion within 30 days; written cert	Yes	Negotiate exit clause; require deletion cert
Availability	99.9% monthly; broad exclusions	99.95% excl. maintenance windows only	Yes	Negotiate exclusion clause limitations
Security/Privacy	72-hour breach notification	12-hour notification per regulator req.	Yes	Negotiate supplemental DPA

Resiliency	Multi-AZ; single-region	Multi-region failover required	Yes	Require multi-region config or compensating control
Performance	Best-effort; no latency SLA	Sub-200ms API response for core path	Yes	Negotiate performance SLA or accept risk
Governance	Customer-managed audit logs; 90-day retention	12-month retention per policy	Yes	Configure extended retention; confirm cost

How to use this: *This matrix should be completed for every cloud service before contract signature and reviewed annually to identify gaps introduced by provider policy changes.*

Knowledge Check

Test your understanding before moving on. Answers are shown below each question.

Q1. A financial institution discovers three months after migrating to a cloud provider that the standard contract permits the provider to retain copies of processed transaction data for eighteen months after termination. Which shared consideration was most critically not evaluated during procurement?

- A) A. Portability, because the data cannot be moved to another provider.
- B) B. Reversibility, because the organisation cannot recover its data and verify the provider no longer holds it after termination.
- C) C. Resiliency, because the provider's data retention creates a single point of failure.
- D) D. Performance, because data retained by the provider may slow down the termination process.

Answer: B. Reversibility, because the organisation cannot recover its data and verify the provider no longer holds it after termination.

Q2. A cloud provider's SLA specifies 99.9% monthly availability. During review, the practitioner discovers that the SLA excludes from downtime calculation all outages caused by 'infrastructure events beyond a single availability zone.' For a workload hosted in a single region, which is the most accurate assessment of this clause?

- A) A. The clause is acceptable because 99.9% availability is an industry-standard commitment.
- B) B. The clause is acceptable because regional outages are rare and the provider's architecture is designed to prevent them.
- C) C. The clause significantly weakens the effective availability commitment because it excludes the scenario most likely to cause extended downtime for a single-region workload.
- D) D. The clause is irrelevant because the customer can deploy to multiple regions to avoid the exclusion.

Answer: C. The clause significantly weakens the effective availability commitment because it excludes the scenario most likely to cause extended downtime for a single-region workload.

Q3. A regulated financial firm discovers its cloud provider's data processing agreement requires breach notification to the customer within 72 hours. The firm's regulatory obligation requires it to notify its regulator within 24 hours of becoming aware of a significant incident. Which action should the cloud security practitioner take?

- A) A. Accept the 72-hour notification term because it aligns with the EU GDPR standard and is therefore compliant.

- B) B. Negotiate a supplemental data processing addendum requiring the provider to notify the firm within a timeframe that allows the firm to meet its own 24-hour regulatory notification obligation.
- C) C. Implement a monitoring capability to detect provider breaches independently, eliminating dependency on the provider's notification obligation.
- D) D. Document the gap in the risk register and accept it as residual risk because large providers cannot accommodate individual customer notification timelines.

Answer: B. Negotiate a supplemental data processing addendum requiring the provider to notify the firm within a timeframe that allows the firm to meet its own 24-hour regulatory notification obligation.

Q4. A cloud security practitioner is evaluating a clinical documentation platform for a hospital trust. The platform's API is proprietary and not documented for external consumption. Which risk does this most directly create?

- A) A. Data sovereignty risk because proprietary APIs may process data in undisclosed locations.
- B) B. Availability risk because undocumented APIs are more likely to experience unexpected downtime.
- C) C. Interoperability risk because the platform cannot integrate with existing systems using standard protocols, likely requiring insecure bridging workarounds.
- D) D. Portability risk because data stored in the platform cannot be exported in standard formats.

Answer: C. Interoperability risk because the platform cannot integrate with existing systems using standard protocols, likely requiring insecure bridging workarounds.

Q5. During a shared considerations evaluation for a new cloud service, the practitioner discovers that the shared responsibility matrix assigns encryption key management to the customer but the operational team believes the provider manages encryption keys. Which risk does this misunderstanding most directly create?

- A) A. Vendor lock-in risk because key management dependency prevents migration.
- B) B. A control gap in which neither party is actively managing encryption keys, leaving data potentially unprotected or protected with provider-default keys that the customer cannot rotate or control.
- C) C. Compliance risk because the contract does not specify which party manages keys.
- D) D. Performance risk because customer-managed key operations add latency to data access.

Answer: B. A control gap in which neither party is actively managing encryption keys, leaving data potentially unprotected or protected with provider-default keys that the customer cannot rotate or control.

Reflection Questions

1. Think about a cloud service your organisation or a client currently uses. Have the shared considerations in this lecture been formally evaluated for that service? Which ones are most likely to have gaps?
2. The financial institution in this lecture's opening story discovered a reversibility problem three months after go-live. What process change would have prevented that discovery from coming as a surprise?
3. SLA exclusion clauses are written by provider legal teams to minimise the provider's exposure. As a cloud security practitioner, how do you build the business case for investing legal resources in SLA negotiation?
4. Security and privacy as shared considerations require contractual commitments from providers on issues like breach notification. How do you approach a situation where the preferred provider refuses to negotiate a key term?
5. Shared responsibility matrices are often provided as high-level diagrams that leave specific allocations ambiguous. How do you verify that the operational team responsible for a cloud service understands which controls they own?

Key Takeaways

- Shared considerations are the contractual and operational foundations of cloud security, not technical configurations, and must be evaluated before service commencement.
- Portability evaluations should include a live data export test using representative data before contract signature, not just a review of provider documentation.
- Reversibility requires explicit contract terms covering data deletion timelines, export format, sub-processor deletion obligations, and written deletion certification.
- SLA availability percentages must be read alongside exclusion clauses; the effective availability commitment after exclusions may be significantly lower than the headline figure.
- Breach notification timelines in provider contracts must be mapped against the customer's own regulatory notification obligations before signature and negotiated where gaps exist.
- Security and privacy shared considerations are contractual obligations that must be reviewed independently of provider certification status; certifications do not address customer-specific regulatory requirements.
- Shared responsibility matrices must be reviewed with both the security team and the operational team for each service, with gaps in control ownership identified and assigned before production use.
- Interoperability weaknesses are commonly bridged by insecure workarounds; requiring standard API protocols and testing integration before commitment prevents this pattern.

Recommended Next Actions

Put this lecture to work within the next seven days:

6. Select one cloud service currently in use and review its contract for exit clause language, specifically checking data deletion timelines, export format commitments, and sub-processor deletion obligations.
7. Review the SLA for one cloud service against the workload's availability requirements, applying each exclusion clause to calculate the effective availability commitment under the most likely failure scenarios.
8. Map your organisation's breach notification obligations across all applicable regulatory frameworks and verify that each cloud provider's data processing agreement supports meeting the most demanding notification deadline.
9. Review the shared responsibility matrix for one cloud service with the operational team responsible for that service and identify any controls where responsibility is ambiguous or assumed to be the other party's obligation.